EXPLAINABLE CLINICAL RISK REPORT — INDIAN LIVER PATIENT DATASET (ILPD)

Patient ID:	PT-2026-001	**Age / Gender:**	19 Yrs / Female
Report Date:	2026-08-05 19:32:54	**Diagnostic Target:**	Indian Liver Patient Dataset (ILPD)

RISK STRATIFICATION SUMMARY	
Predicted Pathological Probability:	**0.7683 (76.8%)**
Assigned Clinical Risk Tier:	**TIER IV - CRITICAL RISK**
Model Prediction Confidence:	53.7%
Clinical Interpretation:	Severe pathological indicators present; immediate specialist referral required.

PATIENT CLINICAL METRICS & NORMAL REFERENCE RANGES

Biomarker	Observed Raw Value	Normal Reference Range
Age	19.00	N/A
Total_Bilirubin	0.70	0.1 - 1.2 mg/dL
Direct_Bilirubin	0.20	0.0 - 0.3 mg/dL
Alkaline_Phosphotase	186.00	44 - 147 IU/L
Alamine_Aminotransferase	166.00	7 - 56 IU/L
Aspartate_Aminotransferase	397.00	10 - 40 IU/L
Total_Protiens	5.50	6.0 - 8.3 g/dL
Albumin	3.00	3.5 - 5.0 g/dL
Albumin_and_Globulin_Ratio	1.20	0.8 - 2.0
Gender	Female	N/A

EXPLAINABLE AI ATTRIBUTIONS (SHAP BREAKDOWN)

Biomarker Factor	Risk Influence Category	SHAP Value (Marginal Impact)
Aspartate_Aminotransferase	Risk Accelerator (+)	+0.1558
Alamine_Aminotransferase	Risk Accelerator (+)	+0.1189

AG_Discrepancy	Risk Accelerator (+)	+0.0599
Gender_Female	Protective Factor (-)	-0.0490
Albumin	Risk Accelerator (+)	+0.0487
Age	Protective Factor (-)	-0.0475

RECOMMENDED FOLLOW-UP CLINICAL PROTOCOLS

1. Confirmatory Testing: Order immediate secondary diagnostic confirmation (e.g. Stress ECG, renal ultrasound, secondary panels).

2. Specialist Referral: Initiate immediate specialist consultation matching the pathological disease profile.

3. Monitoring Frequency: Re-evaluate critical biomarker parameters within 48-72 hours.

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